

Isaac Tilahun

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EDUCATION

University of Toronto | Co-op Internship Program

CGPA: 3.88/4

Honours Bachelor of Science (HBS), Computer Science and Applied Statistics

Expected April 2029

Relevant Coursework: **Operating Systems**, Algorithm Design & Analysis, Databases, Software Design

EXPERIENCE

Attendify

Cambridge, ON

Software Engineer Intern | *PHP, CodeIgniter, MySQL*

Jun 2026 - Sep 2026

- Led **Attendify Wrapped**, a year-end recap for daycare franchises, from architecture proposal to full build: presented the design to the engineering team, prototyped a demo branch, then built the **data model**, a cron job computing **17 stats per franchise** (care hours, billing totals) without modifying source data, and the dashboard in **PHP (CodeIgniter)**.
- Audited controllers, models, and libraries for **SQL injection**, tracing user input to database calls; fixed all high- and medium-severity findings with parameterized queries.
- Extended subscription-renewal invoicing with **automated email delivery**, mark-paid tracking, and overdue detection; refactored it into a mailer library and built the Account Invoices page.
- Built a role-scoped admin activity dashboard with before-and-after field-change logging; fixed the app-wide "Confirm Form Resubmission" error with a **Post-Redirect-Get** shim.

AMD Schola

Toronto, ON

Student Developer | *Python, C++, Godot, Unreal Engine, RL*

Sep 2026 - Present

- Porting **AMD Schola's** reinforcement-learning toolkit from **Unreal Engine** to **Godot** as a **CSC301** capstone project, adapting agent and environment tooling for the new game-engine runtime.
- Evaluating feature parity across **environment creation**, **agent configuration**, and **training-framework integration**, keeping simulation and training logic cleanly separated for future flexibility.
- Collaborating with a **seven-person team** and **two AMD engineers** on weekly porting architecture and implementation decisions.

Code Ninjas

Brampton, ON

Coding Instructor

Apr 2026 - Present

- Teach **100+** students (elementary to Grade 10) programming and game development; designed and led a **12-week intro Python course** from scratch.

PROJECTS

mysh: Unix Shell & Networked Chat | *C, Linux, Systems Programming, Sockets*

Project Details

- Built a **Unix-style shell in C** with variable expansion, custom built-ins, multi-stage pipelines, background jobs, and signal handling using fork, execvp, waitpid, and dup2.
- Extended the shell with **select-based multi-client TCP chat**, multiplexing terminal and socket input without threads and managing active connections through linked lists.

HeyCare | *React, TypeScript, Supabase, Web Speech API, OpenRouter*

GitHub

- Built a **voice-first clinical documentation assistant** in a five-person hackathon team, integrating the Web Speech API, Supabase Edge Functions, and OpenRouter's GPT-4o mini to transform live consultations into structured, editable patient notes.
- Implemented an **end-to-end patient-record workflow** with extraction of 14 clinical fields, speaker-labelled transcripts, Supabase CRUD and multi-field search, fallback handling for failed AI analysis, and TXT/printable HTML exports.

ASL Learning Hub | *JavaScript, MediaPipe, HTML/CSS*

GitHub

- Built a browser-based ASL practice application for A-Z and numbers 1-10, using **MediaPipe Hands to track 21 landmarks** and score gestures in real time.
- Implemented **real-time confidence feedback**, live skeleton overlays, lessons, achievements, and locally persisted progress in a responsive interface.

TECHNICAL SKILLS

- **Languages:** Java, Python, JavaScript, TypeScript, PHP, C, C++, C#, SQL, Bash, Assembly
- **Frameworks & Platforms:** React, Node.js, Express.js, REST APIs, CodeIgniter
- **Databases:** MySQL, PostgreSQL, Supabase
- **Tools:** Git, GitHub, Linux, Claude Code, Codex, Cursor, GitHub Copilot